MERVE DENIZ SENELEN

Frontend Engineer | React, TypeScript, Next.js, Node.js | Berlin, Germany

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Professional Summary

Frontend-focused full-stack engineer with 3+ years of experience in React, TypeScript and Next.js, plus Node.js, PostgreSQL and REST APIs. At Continuous Software, I made a **12,000-row AG Grid table 4x faster** and fixed **60+ UI and accessibility issues**. At BONFRY, I built and owned a frontend from scratch with TDD and CI/CD. MSc in Data Science (2026), focused on machine learning and AI.

Technical Skills

Frontend: React, TypeScript, Redux, Next.js, JavaScript (ES6+), HTML5, CSS3, Sass/SCSS, CSS Modules, Ant Design, Tailwind CSS, Salesforce LWC
Backend: REST APIs, Node.js, PostgreSQL, Fastify, Python (FastAPI), Java (Spring Boot), Zod
Testing & DevOps: Jest, Vitest, Playwright (E2E), TDD, Accessibility (ARIA, keyboard navigation), Git, GitHub Actions (CI/CD), AWS Lambda, Docker, Agile (Scrum, Kanban)
AI & ML: PyTorch, TensorFlow, scikit-learn, pandas, LLM APIs, Claude Code, signal processing, EDA
Languages: English (C2), German (B1), Turkish (native)

Experience

Software Engineer, Working Student (part-time, alongside MSc) *Nov 2025 - Mar 2026*
BONFRY, Berlin, Germany (Hybrid)

- **Frontend Ownership:** Built the frontend from scratch with Salesforce LWC, JavaScript and TDD (Jest), shipping **7 features in 4 months**.
- **End-to-End Delivery:** Owned features from stakeholder planning to release in a small Scrum team, with CI/CD.
- **Data Workflows:** Built **5 AWS Lambda workflows** in Node.js that log every run and reject bad input.
- **Monitoring & Alerts:** Set up Slack alerts that cut the time to spot a failure from hours to **under 1 minute**.

Frontend Developer *Oct 2022 - Sep 2024*
Continuous Software, Dublin, Ireland (Remote)
Main product: Aangine, a strategic portfolio planning platform used by large companies across industries. Worked in a Kanban team.

- **Product Delivery:** Shipped **14 features** in React and TypeScript.
- **Frontend Performance:** Made the largest AG Grid table (**12,000 rows**) load **4x faster**, from 4.5s to 1.1s.
- **Accessibility & UX:** Fixed **60+ UI bugs** and added full keyboard navigation, ARIA labels and focus states.
- **Load Testing:** Found **3 API bottlenecks** by load testing with **300 users** at once.

Software Engineer Intern *Feb 2021 - Oct 2021*
Provera TBT, Adana, Turkey (Remote)

- **Frontend Development:** Built a responsive, cross-browser UI for a project planning web app with React.js, JavaScript and Redux.
- **Backend Support:** Built **18 REST endpoints** in Java Spring Boot with JWT login.

Projects

Melodia | AI melody generator, MSc thesis | melody-generator-ai.vercel.app
Next.js, TypeScript, FastAPI, Python, PyTorch, Vercel

- **Audio AI App:** Web app that turns a short audio clip into a new melody in about **40 seconds**.
- **Generative Model:** Conditional VAE trained on ~**200 hours** of piano music, fine-tuned with reinforcement learning.

OpsLens | Real-time service monitoring dashboard | opslens-sigma.vercel.app
Next.js, TypeScript, Fastify, PostgreSQL, Zod, SSE, Vitest, Playwright, GitHub Actions

- **Real-Time Alerts:** Live alerts every minute. One alert check over **1.38M data points** runs in about **170 ms**.
- **Query Performance:** Measured key queries at 3 data sizes (**6.7 ms** at 13.8k rows, **216.3 ms** at 1.38M) before optimizing.
- **Shared Schemas:** One set of Zod schemas keeps all **6 screens** in sync with the API.
- **Fast & Tested:** Chart library loads on only **1 of 6 pages**; **4 test layers** run in CI before each deploy.

Education

MSc in Data Science, Arden University, Berlin | Sep 2024 - Oct 2026

BSc in Computer Engineering, Adana Science and Technology University, Turkey | Sep 2017 - Jun 2022
Thesis: ALS disease prediction from EMG signals with an LSTM classifier (Keras, TensorFlow).